Global Valuation

Cross-Country Price-Earnings Multiples

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Outline

Data and Sample

Data Sources

File	Size	Rows	Cols	Description
`compustat_global`	2.3 GB	981K	468	Global annual fundamentals (122 countries)
`compustat_america`	0.8 GB	254K	984	North America fundamentals
`returns_global`	1.9 GB	10.4M	25	Global monthly security prices

- **Combined panel**: 743,087 firm-years, 72,720 firms, 81 countries (1986–2024)
- All financial data in *local currency* (millions)
- **Market cap (Global)**: `cshoi` × `prccm` using `prirow` (primary security)

Variable Definitions

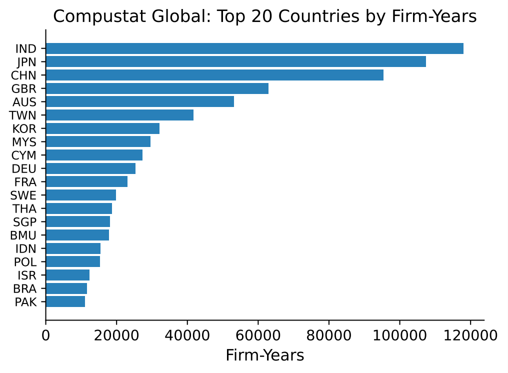
Dependent variable:

$$\log(P/E)_{i,t} = \log\left(\frac{\text{Market Cap}_{i,t}}{\text{Net Income}_{i,t}}\right), \quad NI > 0$$

Market Cap	US: `mkvalt`. Global: `cshoi` $\times$ `prccm` (primary security)
Net Income	`ib` (income before extraordinary items)
ROA	`ib` / `at`
Leverage	(`at` $-$ `ceq`)/`at`
Industry	GICS Industry Group ($\sim$25 categories)

P/E restricted to profitable firms (72% of sample). All ratios winsorized at 1st/99th percentiles.

Coverage and Descriptive Statistics

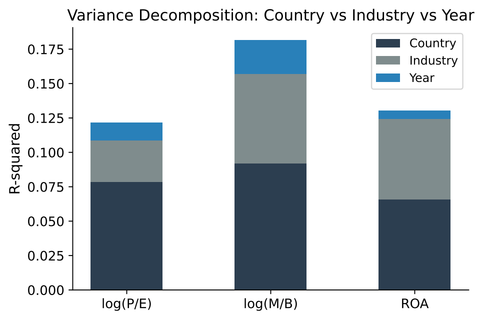


Variable	N	Median	P10	P90
log(P/E)	532K	2.79	1.76	4.32
ROA	743K	0.02	−0.15	0.11
Leverage	743K	0.49	0.14	0.85

Top 5 by obs: USA (188K), JPN (60K), CHN (58K), IND (57K), TWN (31K). Global prices start 2007.

Country Dominates P/E

Variance Decomposition: Country vs Industry vs Year



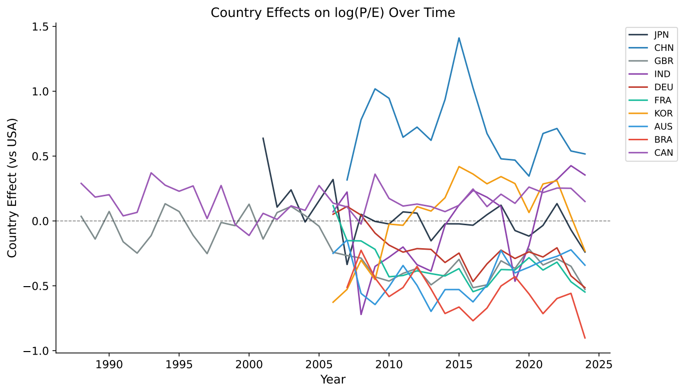
Shapley R^2 decomposition of log(P/E):

Factor	Share of R^2
Country	**64.6%**
Industry	24.8%
Year	10.6%

Country alone explains nearly two-thirds of the variation in P/E.

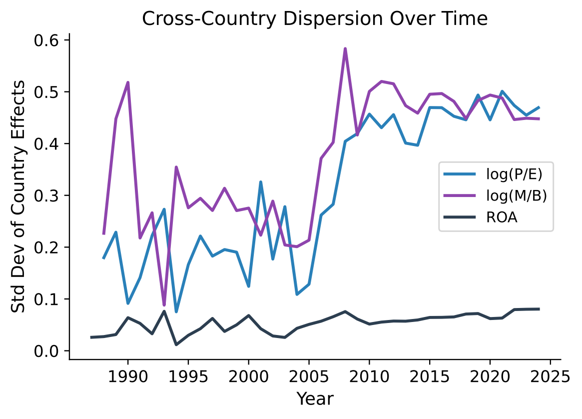
Extends Heston–Rouwenhorst (1994) from *returns* to *valuation levels*.

Country P/E Effects Over Time (vs USA)



Year-by-year regressions with size, leverage, earnings growth, industry FE. USA = 0. China premium largest but declining. Japan near zero. India rising since 2015.

P/E Dispersion Is Diverging



P/E dispersion:

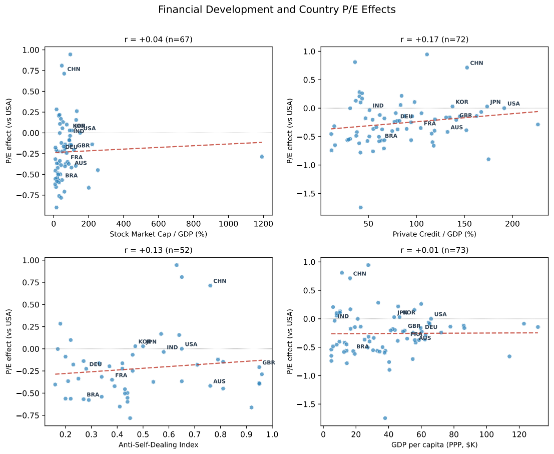
$$\sigma = 0.17 \;\rightarrow\; 0.47 \quad (\times 2.7)$$

Despite financial globalization, country valuations are *not converging*.

The gap between the most and least expensive countries has nearly tripled since the early 2000s.

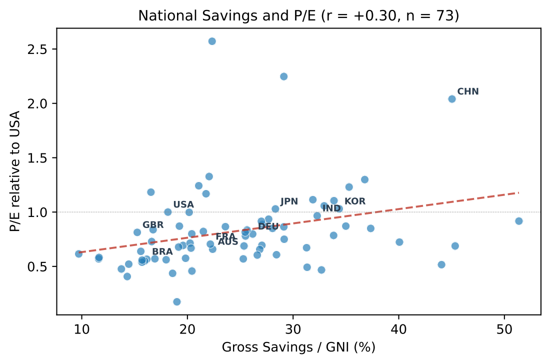
What Explains Cross-Country P/E?

Standard Institutional Variables Do Not Predict P/E



Market cap/GDP ($r = +0.04$), credit/GDP ($r = +0.17$), anti-self-dealing ($r = +0.13$), GDP per capita ($r = +0.01$): all near zero. La Porta et al. (2002) variables do not explain valuation levels.

National Savings Rate Predicts P/E



Gross savings / GNI (World Bank):

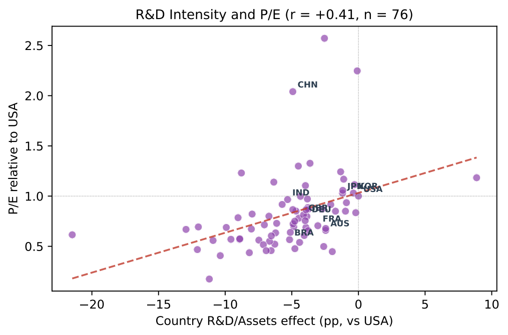
$r = +0.30, \quad t = 2.65^{***} \quad (n = 73)$

Robust:

- Excl. Gulf states: $t = 3.40$
- Controlling for R&D: $t = 2.75$
- Controlling for GDP growth: $t = 2.88$

Demand-side channel: more capital seeking equity returns pushes up price multiples.

R&D Intensity Predicts P/E

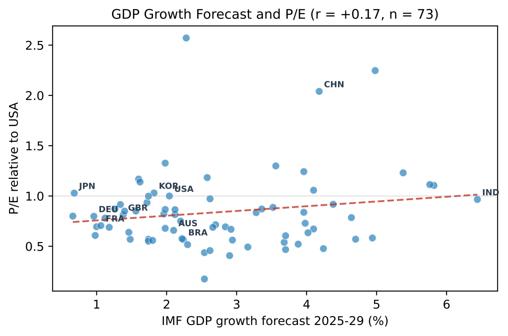


Country R&D/Assets vs P/E: $r = +0.41$, $t = 3.91$

Physical investment (CapEx): null ($r = +0.03$). Not an accounting artifact: adding R&D back to earnings does not change the cross-country pattern. Not driven by IFRS vs GAAP differences.

Supply-side channel: high-P/E countries invest more in *intangibles*, not physical capital.

Not About Expected Growth



IMF 5-year GDP growth forecasts vs P/E:
$r = +0.17, \quad t = 1.42 \quad$ (null)

Also null: realized earnings growth ($r \approx 0$), realized GDP growth ($r = +0.18$, $t = 1.56$).

Cross-country P/E differences are **not** driven by expected growth rates. India (6.4% forecast) has P/E $\approx$ USA. Japan (0.7%) matches Korea (1.8%).

Growth becomes marginally significant ($t = 1.69$) only as a third control alongside R&D and savings.

Savings and R&D: Two Independent Channels

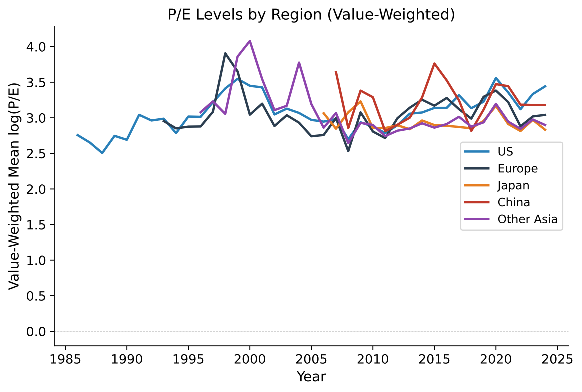
P/E $\sim$	R&D	Savings	Growth	R^2
R&D only	4.86 (4.73)			0.23
Savings only		0.016 (3.10)		0.12
R&D + Savings	4.67 (4.72)	0.013 (2.75)		0.35
All three	**6.29 (5.89)**	**0.014 (2.88)**	0.051 (1.69)	**0.45**

R&D and savings both survive jointly. Growth is a useful control but not independently significant. Together: $R^2 = 0.45$ ($n = 65$).

55% of cross-country P/E variation remains unexplained — large residual country effects persist (e.g. China, Chile).

Regional Patterns

Regional P/E Over Time

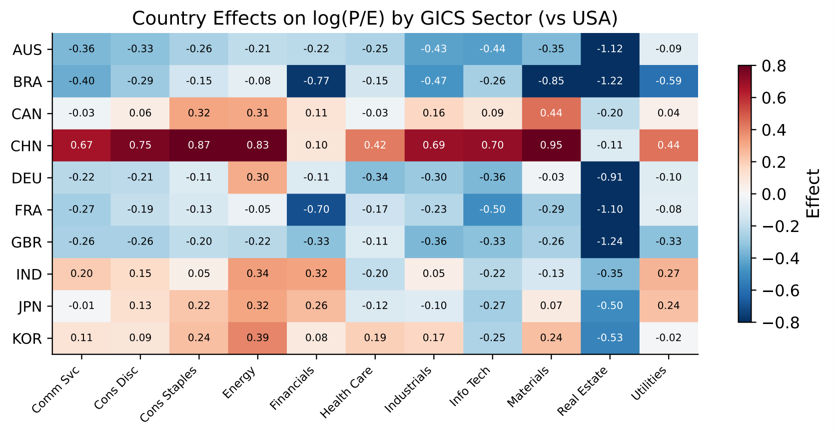


Key patterns:

- US P/E premium is recent and driven by large firms (tech)
- China P/E volatile, declining since 2015
- Japan near Europe

Value-weighted averages. VW gaps larger than EW — US premium concentrated in mega-caps.

Industry × Country P/E Effects



Country P/E effects by GICS sector. The China premium is broad-based across industries. Real estate shows large discounts in most countries.

Summary and Next Steps

Stylized Facts

1. **Country dominates P/E** — 65% of R^2 (Shapley). Extends Heston–Rouwenhorst (1994) from returns to valuation levels.
2. **P/E dispersion is diverging** — $\times 2.7$ since early 2000s, despite financial globalization.
3. **Two robust predictors of cross-country P/E:**
   - **R&D intensity** ($r = +0.41$, $t = 3.91$) — supply-side: firms investing in intangibles.
   - **National savings rate** ($r = +0.30$, $t = 2.65$) — demand-side: more capital chasing equities.

   Together with growth controls: $R^2 = 0.45$. Both survive jointly.
4. **Not about expected growth** — IMF GDP forecasts, realized earnings growth: all insignificant. Institutional variables (La Porta), pension funds, equity depth: also null.

Research Questions

1. **Why is P/E dispersion diverging?** ($\times 2.7$) — US tech? Post-GFC monetary policy? Capital flows?
2. **Can the savings channel be formalized?** — Savings/GNI predicts P/E and is orthogonal to ROA. A model where savings rates affect discount rates?
3. **Do low-P/E countries earn higher returns?** — Bekaert–Harvey, Fama–French international. If so, valuation gaps reflect risk premia.
4. **What explains China's P/E premium?** — Highest P/E among large countries. Retail investors? Capital controls? Savings rate (46% of GNI)?

Related Literature

Country vs industry effects:

- Heston–Rouwenhorst (1994, *JFE*) — country dominates returns. We extend to *levels*.
- Griffin–Karolyi (1998, *JFE*) — industry effects vary across sectors.

Institutions and investor protection:

- La Porta et al. (2002, *JF*) — investor protection raises Tobin's Q. Our results: weak for P/E.
- Hail–Leuz (2006, *JAR*) — institutions reduce cost of equity by up to 200 bps.
- Stulz (2005, *JF*) — "twin agency problems" prevent equalization.

Savings and asset prices:

- Abel (2003, *Brookings*) — effect of baby boomers' saving on asset prices.
- Caballero et al. (2008, *JME*) — global imbalances and asset price inflation.

Appendix

P/E Regression Specification

Main specification (profitable firms, $NI > 0$):

$$\log(P/E)_{i,t} = \beta_1 \text{Lev} + \beta_2 \log(\text{Assets}) + \beta_3 \Delta E_{t-1} + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}$$

	Spec 1	Spec 2
Leverage	-0.017 (0.102)	-0.013 (0.100)
log(At)	-0.076** (0.030)	-0.078** (0.031)
Lag ΔE	-0.016*** (0.003)	-0.016*** (0.003)
FE	Country + Ind + Year	Country×Year + Ind
N / R^2	428K / 0.142	428K / 0.163

SE clustered by country. Year-by-year regressions extract γ_c with USA $= 0$.

Full List of Tested Predictors

Variable	r with P/E	t-stat	N
R&D / Assets	**+0.48*** **	**4.73**	76
Savings / GNI	**+0.34*** **	**3.13**	68
ROA	−0.39	−3.75	76
after R&D control	*−0.08*	*−0.76*	76
Private Credit / GDP	+0.17	1.30	72
Pension Fund Assets / GDP	+0.14	1.16	56
IMF GDP growth forecast	+0.13	1.07	73
Anti-Self-Dealing Index	+0.13	0.92	52
Stock Market Cap / GDP	+0.04	0.29	67
GDP per capita (PPP)	+0.01	0.09	73
Realized earnings growth	−0.00	−0.04	80
CapEx / Assets	+0.03	0.24	78

ROA appears significant alone but is absorbed by R&D. The ROA–P/E correlation is mediated by intangible investment intensity.

Market Cap Construction (Detail)

US firms: `mkvalt` (pre-computed market value in USD millions). Available from 1986.

Non-US firms (Compustat Global): no pre-computed market value.

- Compute: Market Cap = `cshoi` (shares outstanding) $\times$ `prccm` (month-end price)
- **Pitfall**: firms have multiple securities across exchanges and currencies

  Example — SAP SE (gvkey=103487) in Dec 2022:

iid=02W	EUR	96.39	← correct
iid=05W	HUF	38,620	← wrong currency!
iid=06W	MXN	2,009	← wrong currency!

- **Solution**: use `prirow` field (primary security designation)

All data in local currency (millions). Cross-country *levels* absorbed by country FE.

Accounting Standards

Compustat Global reports accounting standard via `acctstd`:

Standard	**Share**
IFRS	59.7%
Domestic GAAP	39.3%
US GAAP	0.2%
Other / missing	0.8%

US firms: all US GAAP.

Country variation:

- EU: mandatory IFRS since 2005
- Japan: 93% domestic GAAP
- China: 96% domestic GAAP
- Korea: 99% IFRS (since 2011)

P/E is less sensitive to accounting standards than book-equity-based ratios. Country FE absorb time-invariant differences.